Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/03/2006
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Role of the Interleukin-2 Receptor in Signal Transduction		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Northern California Institute for Research and Education		
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PI Address 2:	4150 Clement Street		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	01/15/1999	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Soyuz/ISS		
Flight Assignment:	Soyuz (changed 7/07 per PI) ISS (changed 7/07 per PI)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cogoli, Augusto (Swiss Federal Institute of Technology)		
Grant/Contract No.:	NAG2-1286		
Performance Goal No.:			
Performance Goal Text:			
	This work is collaboration between Professor Augusto Cogoli and Professor Millie Hughes-Fulford to study the role of the interleukin-2 receptor in signal transduction of T cells. The nearly total loss of T cell activation in real microgravity was discovered in a Spacelab in 1983 and was confirmed later by several experiments in space. Inhibition of IL-2R expression is probably one of the causes of such loss of activation. It is planned to analyze the pathway of mitogenic signal transduction in T cells in true microgravity as well as at variable G levels between 0 and 1xg. The objectives of this research project are: 1. To investigate selected critical steps of T cell activation 2. To test the hypothesis that a failure in the expression of the IL-2R is causing the loss of activity in microgravity 3. To identify windows of sensitivity to gravity during the activation process 4. To establish thresholds of gravi-sensing T cells.		

Task Description:	The main experimental approach will consist of the activation of cultures of purified peripheral blood cells with T cell mitogens in real and simulated microgravity. The specific IL-2R mRNA will be quantitatively determined with the reverse Transcriptase-polymerase chain reaction (RT-PCR) technology, the insertion of IL-2R in the membrane will be visualized by immunofluorscence and its secretion in the supernatant will be measured by immunoassay. Signal transduction will be furthered analyzed by determining the early expression of oncogenes. The Laboratory of Cell Growth (LCG), headed by Dr. Millie Hughes-Fulford is responsible for analysis of gene expression in T cells. The nearly total loss of T cell activation in real microgravity was discovered in a Spacelab in 1983 and was confirmed later by several experiments in space. Using gene chip technology we have found new pathways that regulate early T cell activation and these are in press. Using the RPM, we are able to simulate microgravity and define which pathways are dysfunctional in altered gravity, and hence specifics of gravity based life.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	All terrestrial life began in a gravity field. This study of alterations of early T-cell activation microgravity gives us the unique opportunity to examine the role of Earth's gravity in immune function. In the late 60's returning Apollo astronauts were found to have reduced immune function after spaceflight, taking approximately 7 days to recover normal function. Later studies by Dr. Cogoli demonstrated changes in the T-cells during exposure to microgravity. This program will examine the expression of IL-2, and the three IL-2 receptor subunits (alpha, beta and gamma) in microgravity with and without an artificial 1g gravity vector and compare those results with ground controls. These studies will show the regulation of the earliest signals that cause the T-cell to activate and the role of normal earth's gravity in that signaling.
Task Progress:	We have made significant progress this year in definition of the flight experiment and developing protocols for the flight. No funding was received for the 2005 year. This experiment will fly on Soyuz to the ISS flight estimated for September, 2006. We have developed our reatime rtpcr techniques in this laboratory and are able to verify chip data with qrtpcr. We have found T-cell activation is altered in a simulated microgravity situation. We have also found Changes in T-cell gene expression due to lack of gravity vector. Data from Affymetrix gene chips will be used to analyze the patterns and clustering of gene expression in the flight experiment. Using Affy Mass5 and GeneSpring we have analyzed gene expression from T-cells that were mitogen activated in a normal 1-g vector or in a random g-vector (random positioning machine, RPM). we have compared the gene expression of cells activated in a 1-g vector or in the RPM with no g-vector (B). The gene expression in A and B are different, both in abundance and in genes expressed. Most striking is that in the gravity vector, over 168 genes are induced over 4 fold within 4 hours of activation, while only 17 of those genes are induced the RPM activated samples. We have also started clustering the data to examine both the pattern of expression as well as beginning to examine the specific pathways that are induced by gravity. n=6 for each condition. These data have been published in high impact peer reviewed journals.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Hughes-Fulford M, Rodenacker K, Jutting U. " Reduction of anabolic signals and alteration of osteoblast nuclear morphologyin microgravity." J Cell Biochem. 2006 Apr 17; Epub ahead of print. PMID: 16619267 , Apr-2006
Articles in Peer-reviewed Journals	Hughes-Fulford M, Li CF, Boonyaratanakornkit J, Sayyah S. "Arachidonic acid activates phosphatidylinositol 3-kinase signaling and induces gene expression in prostate cancer." Cancer Res. 2006 Feb 1;66(3):1427-33. PMID: 16452198 , Feb-2006
Articles in Peer-reviewed Journals	Boonyaratanakornkit JB, Cogoli A, Li CF, Schopper T, Pippia P, Galleri G, Meloni MA, Hughes-Fulford. "Key gravity-sensitive signaling pathways drive T cell activation." FASEB J. 2005 Dec;19(14):2020-2. PMID: 16210397 , Dec-2005
Articles in Peer-reviewed Journals	Hughes-Fulford M, Tjandrawinata RR, Li CF, Sayyah S. "Arachidonic acid, an omega-6 fatty acid, induces cytoplasmic phospholipase A2 in prostate carcinoma cells." Carcinogenesis. 2005 Sep;26(9):1520-6. PMID: 15878913 , Sep-2005
Articles in Peer-reviewed Journals	Hughes-Fulford M, Sugano E, Schopper T, Li CF, Boonyaratanakornkit JB, Cogoli A. "Early immune response and regulation of IL-2 receptor subunits." Cell Signal. 2005 Sep;17(9):1111-24. PMID: 15993752 , Sep-2005